6 Figure 6.1 is a full-scale drawing of an object O, a thin converging lens L and the image I of object O.

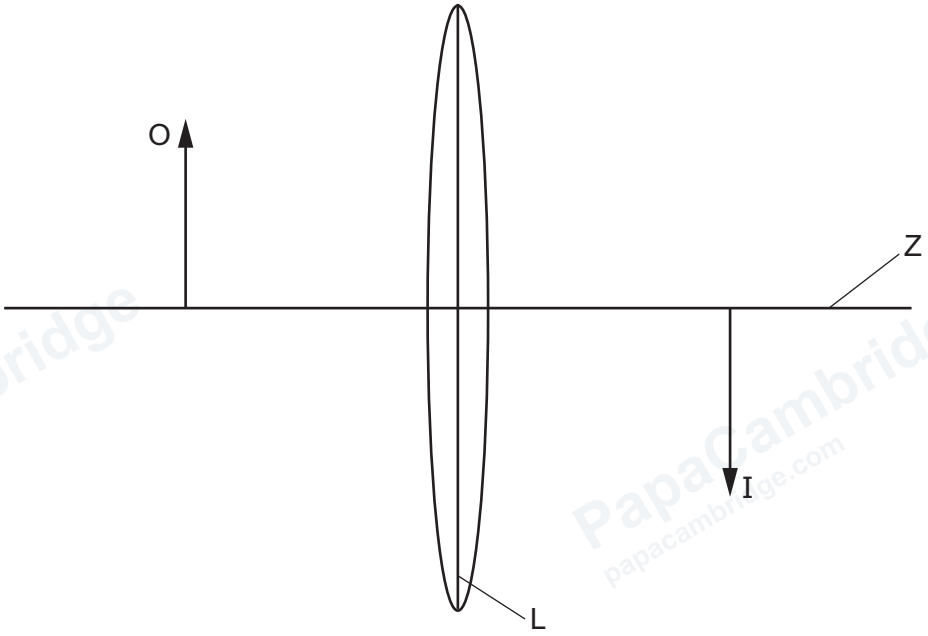

Figure 6.1

(a) (i) Identify Z.

..... [1]

(ii) Determine the focal length of the lens L. Show any working in the space provided or on Figure 6.1.

focal length = ... cm [2]

(iii) The image in Figure 6.1 is inverted.

State **two** other characteristics of the image.

1 .....

2 .....

[2]

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(b) Figure 6.2 shows a different converging lens with a focal length F.

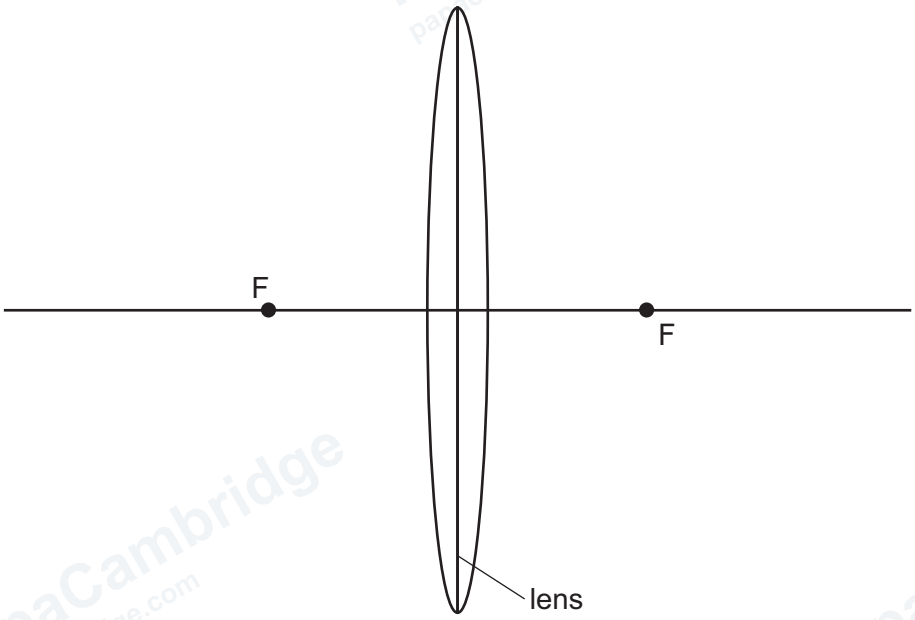

Figure 6.2

(i) The lens in Figure 6.2 is used as a magnifying glass.

Draw an X on Figure 6.2, to show a position for an object which produces a magnified image. [1]

(ii) Put a tick (✓) alongside the **two** statements which describe the position of the magnified image.

on the opposite side of the lens to the object	
on the same side of the lens as the object	
the same distance from the lens as the object.	
closer to the lens than the object	
further away from the lens than the object	

[1]

[Total: 7]

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